PASSAGE: A Multi-Resolution Elevation-Based Pathfinding Dataset for Certified Machine Learning Systems

Author One*
Affiliation One
author1@example.com
& *AuthorTwo**
Affiliation Two
author2@example.com
& *AuthorThree*
Affiliation Three
author3@example.com

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Abstract

We present PASSAGE, a large-scale, multi-resolution pathfinding benchmark dataset derived from real-world elevation data (JAXA ALOS AW3D30). Unlike existing grid-based benchmarks with binary obstacles, PASSAGE features continuous elevation-based cost maps, procedurally generated superellipse obstacles, and ground truth paths from multiple solver backends. The dataset spans seven resolution levels (64×64 to 4096×4096 pixels) with over 13,000 samples across stratified train/calibration/validation/test splits. We provide five cost models (elevation, distance, slope-uphill, energy, absolute-elevation) and three solver backends (NetworkX, pathfinding, NumPy-grid). Designed with aerospace certification standards (ARP6983, ED-324) in mind, PASSAGE offers a bounded Operational Design Domain (ODD) suitable for developing certifiable ML-based pathfinding systems. We release the full generation toolkit and dataset on Hugging Face Hub to facilitate reproducible research in surrogate modeling, reinforcement learning, and graph neural networks for pathfinding.

1 Introduction

1.1 Context and Motivation

The rise of learning-based approaches for pathfinding—including surrogate models, reinforcement learning agents, and graph neural networks—has created demand for standardized benchmark datasets. However, existing benchmarks suffer from several limitations: binary obstacle grids lack the continuous cost gradients found in real terrain, fixed resolutions prevent scalability studies, and limited sample counts hinder modern deep learning.

*Equal contribution.

Simultaneously, safety-critical domains such as aerospace and autonomous vehicles require certifiable AI systems with well-defined Operational Design Domains (ODD). Standards like ARP6983 and ED-324 mandate reproducible test conditions and bounded performance metrics—requirements that current pathfinding datasets do not address.

1.2 Contributions

We make the following contributions:

1. **PASSAGE Dataset**: A large-scale, multi-resolution pathfinding benchmark with 13,880+ samples across seven resolutions (64×64 to 4096×4096).
2. **Procedural Obstacle Generation**: Configurable superellipse-based obstacles with rich shape variety (Lamé curves with $n \in [1, 80]$).
3. **Multi-Backend Solver Framework**: Three solver backends (NetworkX, pathfinding, NumPy-grid) with five cost models.
4. **Certification-Oriented Design**: Bounded ODD, reproducible pipeline, and alignment with aerospace standards.
5. **Open-Source Toolkit**: Complete pipeline from download to Hugging Face Hub export.

2 Related Work

2.1 Existing Pathfinding Datasets

Grid-Based Benchmarks. The Moving AI Lab benchmarks Sturtevant (2012) provide maps from games (Dragon Age, StarCraft) but feature binary obstacles and fixed resolutions. Random grid environments lack terrain realism.

Continuous Environments. Robot navigation datasets and autonomous driving benchmarks (nuScenes, Waymo) operate in continuous space with sensor dependencies, making them unsuitable for grid-based algorithm evaluation.

Elevation-Based Data. Digital Elevation Models (DEMs) are available for GIS applications but lack pathfinding ground truth and standardized ML splits.

2.2 Learning-Based Pathfinding

Supervised Approaches. U-Net architectures predict path masks Ronneberger et al. (2015), while Value Iteration Networks Tamar et al. (2016) embed planning in neural networks. Neural A* Yonetani et al. (2021) learns heuristics for classical solvers.

Reinforcement Learning. DQN and policy gradient methods have been applied to navigation Mirowski et al. (2017), though sample efficiency and generalization remain challenges.

Graph Neural Networks. GNNs naturally represent grid connectivity and have shown promise for combinatorial optimization Cappart et al. (2023).

2.3 Certification Standards

ARP6983 SAE International (2024) and ED-324 EUROCAE WG-114 (2024) provide guidelines for certifying ML systems in aviation, requiring defined ODDs, data quality assurance, and performance monitoring.

3 Dataset Design

3.1 Data Source

PASSAGE uses the JAXA ALOS AW3D30 Digital Surface Model Tadono et al. (2014):

- **Resolution**: 30-meter ground sampling distance
- **Coverage**: Near-global (±60 latitude)
- **Format**: GeoTIFF tiles, 3600×3600 pixels per tile

3.2 Multi-Resolution Support

We generate samples at seven resolution levels (Table 1), enabling studies of computational scalability and resolution-dependent performance.

Table 1. Resolution levels and sample counts in PASSAGE.

Resolution	Samples	Grid Cells	Complexity
64×64	10,000	4,096	Lightweight
128×128	2,500	16,384	Small
256×256	1,000	65,536	Medium
512×512	250	262,144	Large
1024×1024	100	1,048,576	Heavy
2048×2048	20	4,194,304	Very Heavy
4096×4096	10	16,777,216	Extreme

3.3 Obstacle Generation

Obstacles are generated using superellipses (Lamé curves):

$$\left(\frac{|x - c_x|}{a}\right)^n + \left(\frac{|y - c_y|}{b}\right)^n \leq 1 \tag{1}$$

where n controls shape ($n = 1$: diamond, $n = 2$: ellipse, $n > 2$: squarish). Parameters are sampled from configurable distributions (log-uniform default).

3.4 Cost Models

We provide five cost models for edge weights:

$$w_{\text{elev}} = d \cdot (1 + \alpha \cdot \bar{e}_{\text{dst}}) \tag{2}$$

$$w_{\text{dist}} = d \tag{3}$$

$$w_{\text{slope}} = d \cdot (1 + \alpha \cdot \max(0, \Delta e)/\Delta e_{\text{range}}) \tag{4}$$

$$w_{\text{energy}} = d + \alpha \cdot \max(0, \Delta e) \tag{5}$$

$$w_{\text{abs}} = d \cdot (1 + \alpha \cdot |\bar{e}_{\text{dst}}|) \tag{6}$$

where d is physical distance (30m cardinal, 42.43m diagonal), α is a configurable weight, and $\bar{e}$ denotes normalized elevation.

4 Experiments

4.1 Solver Comparison

We benchmark three solver backends across resolutions (Figure **??**).

4.2 Baseline Models

We evaluate baseline architectures:

- **U-Net**: Path segmentation (pixel accuracy, IoU)
- **GNN**: Node classification on graph representation
- **RL (PPO)**: Navigation policy learning

5 Conclusion

We introduced PASSAGE, a multi-resolution elevation-based pathfinding dataset designed for certifiable ML systems. With over 13,000 samples, five cost models, and alignment with aerospace certification standards, PASSAGE enables rigorous benchmarking of surrogate models, RL agents, and GNNs. Future work includes waypoint support, multi-agent scenarios, and temporal dynamics.

Acknowledgments

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A Dataset Configuration Schema

Full configuration schema available in supplementary materials.

B Additional Experiments

Extended results and ablation studies.